

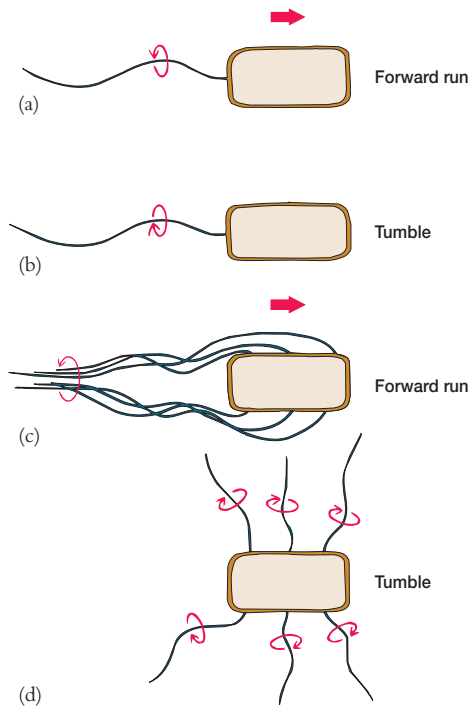


Figure 3.35 Flagellar Motility. The relationship of flagellar rotation to bacterial movement. Parts (a) and (b) describe the motion of monotrichous, polar bacteria. Parts (c) and (d) illustrate the movements of peritrichous organisms.

body in gram-negative bacteria. The rotor portion of the motor seems to be made primarily of a rod, the M ring, and a C ring joined to it on the cytoplasmic side of the basal body. These two rings are made of several proteins; Fli G is particularly important in generating flagellar rotation. The two most important proteins in the stator part of the motor are Mot A and Mot B. These form a proton channel through the plasma membrane, and Mot B also anchors the Mot complex to cell wall peptidoglycan. There is some evidence that Mot A and Fli G directly interact during flagellar rotation. This rotation is driven by proton or sodium gradients in procaryotes, not directly by ATP as is the case with eucaryotic flagella.

The flagellum is a very effective swimming device. From the bacterium's point of view, swimming is quite a task because the surrounding water seems as thick and viscous as molasses. The cell must bore through the water with its helical or corkscrew-shaped flagella, and if flagellar activity ceases, it stops almost instantly. Despite such environmental resistance to movement, bacteria can swim from 20 to almost 90 $\mu\text{m}/\text{second}$. This is equivalent to traveling from 2 to over 100 cell lengths per second. In contrast, an exceptionally fast 6 ft human might be able to run around 5 body lengths per second.

Bacteria can move by mechanisms other than flagellar rotation. Spirochetes are helical bacteria that travel through viscous substances such as mucus or mud by flexing and spinning movements caused by a special **axial filament** composed of periplas-

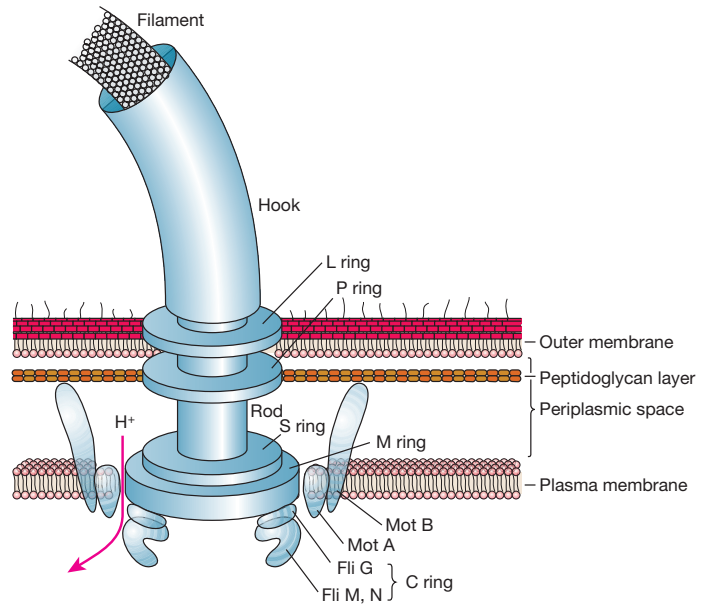


Figure 3.36 Mechanism of Flagellar Movement. This diagram of a gram-negative flagellum shows some of the more important components and the flow of protons that drives rotation. Five of the many flagellar proteins are labeled (Mot A, Mot B, Fli G, Fli M, Fli N).

mic flagella (see section 21.6). A very different type of motility, **gliding motility**, is employed by many bacteria: cyanobacteria (see section 21.3), myxobacteria (see section 22.4) and cytophagas (see section 21.7), and some mycoplasmas (see section 23.1). Although there are no visible external structures associated with gliding motility, these bacteria can coast along solid surfaces at rates up to 3 $\mu\text{m}/\text{second}$. [The mechanism of gliding motility \(Box 21.1\)](#)

1. Briefly describe capsules, slime layers, glycocalyxes, and S-layers. What are their functions?
2. Distinguish between fimbriae and sex pili, and give the function of each.
3. Be able to discuss the following: flagella distribution patterns, flagella structure and synthesis, and the way in which flagella operate to move a bacterium.

3.7 Chemotaxis

Bacteria do not always swim aimlessly but are attracted by such nutrients as sugars and amino acids, and are repelled by many harmful substances and bacterial waste products. (Bacteria also can respond to other environmental cues such as temperature,

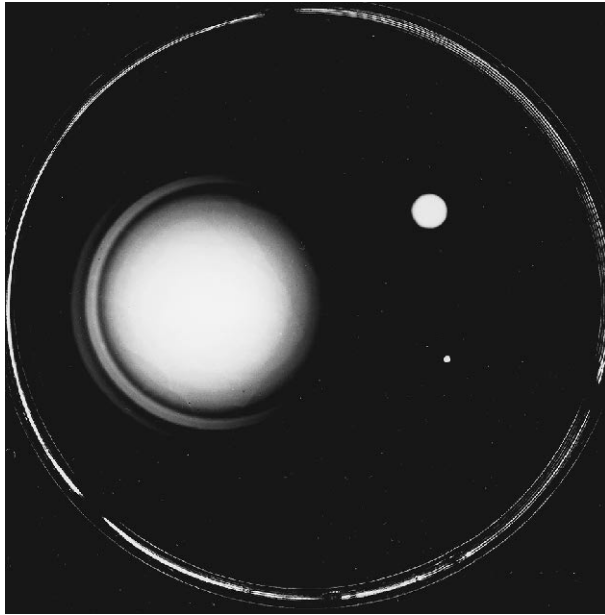


Figure 3.37 Positive Bacterial Chemotaxis. Chemotaxis can be demonstrated on an agar plate that contains various nutrients. Positive chemotaxis by *Escherichia coli* on the left. The outer ring is composed of bacteria consuming serine. The second ring was formed by *E. coli* consuming aspartate, a less powerful attractant. The upper right colony is composed of motile, but nonchemotactic mutants. The bottom right colony is formed by nonmotile bacteria.



Figure 3.38 Negative Bacterial Chemotaxis. Negative chemotaxis by *E. coli* in response to the repellent acetate. The bright disks are plugs of concentrated agar containing acetate that have been placed in dilute agar inoculated with *E. coli*. Acetate concentration increases from zero at the top right to 3 M at top left. Note the increasing size of bacteria-free zones with increasing acetate. The bacteria have migrated for 30 minutes.

light, and gravity; Box 3.3.) Movement toward chemical attractants and away from repellents is known as **chemotaxis**. Such behavior is of obvious advantage to bacteria.

Chemotaxis may be demonstrated by observing bacteria in the chemical gradient produced when a thin capillary tube is filled with an attractant and lowered into a bacterial suspension. As the attractant diffuses from the end of the capillary, bacteria collect and swim up the tube. The number of bacteria within the capillary after a short length of time reflects the strength of attraction and rate of chemotaxis. Positive and negative chemotaxis also can be studied with petri dish cultures (**figure 3.37**). If bacteria are placed in the center of a dish of agar containing an attractant, the bacteria will exhaust the local supply and then swim outward following the attractant gradient they have created. The result is an expanding ring of bacteria. When a disk of repellent is placed in a petri dish of semisolid agar and bacteria, the bacteria will swim away from the repellent, creating a clear zone around the disk (**figure 3.38**).

Bacteria can respond to very low levels of attractants (about 10^{-8} M for some sugars), the magnitude of their response increasing with attractant concentration. Usually they sense repellents only at higher concentrations. If an attractant and a repellent are present together, the bacterium will compare both signals and respond to the chemical with the most effective concentration.

Attractants and repellents are detected by **chemoreceptors**, special proteins that bind chemicals and transmit signals to the

other components of the chemosensing system. About 20 attractant chemoreceptors and 10 chemoreceptors for repellents have been discovered thus far. These chemoreceptor proteins may be located in the periplasmic space or the plasma membrane. Some receptors participate in the initial stages of sugar transport into the cell.

The chemotactic behavior of bacteria has been studied using the tracking microscope, a microscope with a moving stage that automatically keeps an individual bacterium in view. In the absence of a chemical gradient, *E. coli* and other bacteria move randomly. A bacterium travels in a straight or slightly curved line, a **run**, for a few seconds; then it will stop and **tumble** or **twiddle** about. The tumble is followed by a run in a different direction (**figure 3.39**). When the bacterium is exposed to an attractant gradient, it tumbles less frequently (or has longer runs) when traveling up the gradient, but tumbles at normal frequency if moving down the gradient. Consequently the bacterium moves up the gradient. Behavior is shaped by temporal changes in chemical concentration: the bacterium compares its current environment with that experienced a few moments previously; if the attractant concentration is higher, tumbling is suppressed and the run is longer. The opposite response occurs with a repellent gradient. Tumbling frequency decreases (the run lengthens) when the bacterium moves down the gradient away from the repellent.

Although bacterial chemotaxis appears to be deliberate, directed movement, it is important to keep in mind that this is not

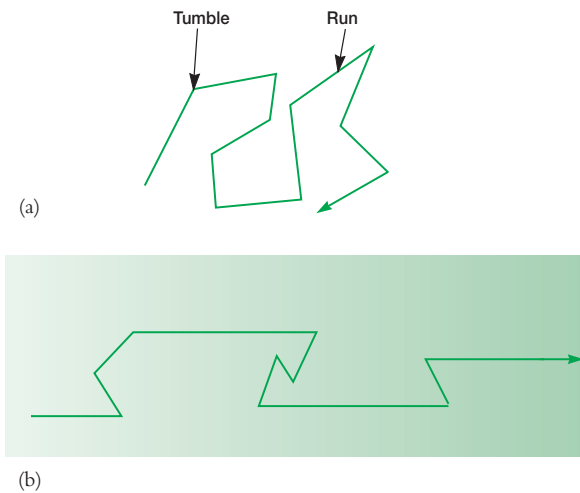


Figure 3.39 Directed Movement in Bacteria. (a) Random movement of a bacterium in the absence of a concentration gradient. Tumbling frequency is fairly constant. (b) Movement in an attractant gradient. Tumbling frequency is reduced when the bacterium is moving up the gradient. Therefore runs in the direction of increasing attractant are longer.

the case. When the environment is constant, bacteria tend to move in a random walk. That is, there is a random sequence of runs followed by tumbles. If a run is in the direction of improving conditions, tumbles are suppressed so that the cell tends to move in the preferred direction. This is said to be a biased random walk toward attractants and away from repellants. Individual cells do not choose a particular direction. Instead, they determine whether or not to continue in the same direction.

Much work has been done on the mechanism of chemotaxis in *Escherichia coli*. Recall that forward swimming is due to counterclockwise rotation of the flagellum, whereas tumbling results from clockwise rotation. The bacteria must be able to respond to gradients in such a way that they collect in nutrient-rich regions and at the proper oxygen level while avoiding toxic materials. *E. coli* has four different chemoreceptors that recognize serine, aspartate and maltose, ribose and galactose, and dipeptides, respectively. These chemoreceptors often are called methyl-accepting chemotaxis proteins (MCPs). They seem to be localized in patches, often at the end of a rod-shaped cell like *E. coli*. The MCPs do not directly influence flagellar rotation but act through a series of proteins. The whole process is so efficient that a stimulus can trigger a motor response in less than 200 milliseconds.

The molecular mechanism underlying chemotaxis is quite complex. It involves conformational changes in proteins, protein methylation, and protein phosphorylation. When an attractant such as a nutrient molecule is not bound to an MCP, the CheA protein is phosphorylated using ATP. This phosphorylated protein can donate its phosphate to the CheY protein, which then interacts with the FlhM switch protein at the base of

the flagellum to promote clockwise rotation and tumbling. An increase in nutrient binding will lead to dephosphorylated CheA, counterclockwise flagellar rotation, and a run. When no attractants or repellants are present, the system maintains intermediate levels of CheA phosphate and CheY phosphate. This produces a normal random walk pattern. In very general terms, the system has a sensory protein that can be phosphorylated and then phosphorylate another protein to cause a response. As we shall see later, this is called a two-component phosphorelay system. The molecular details of the chemotaxis system will be described in chapter 12 when two-component systems are discussed. It should be noted here that a similar mechanism is used to respond to other environmental factors such as oxygen (aerotaxis), light (phototaxis), temperature (thermotaxis), and osmotic pressure (osmotaxis). [Two-component phosphorelay systems \(pp. 283–85\)](#)

1. Define chemotaxis, run, and tumble or twiddle.
2. Explain in a general way how bacteria are attracted to substances like nutrients while being repelled by toxic materials.

3.8 The Bacterial Endospore

A number of gram-positive bacteria can form a special resistant, dormant structure called an **endospore**. Endospores develop within vegetative bacterial cells of several genera: *Bacillus* and *Clostridium* (rods), *Sporosarcina* (cocci), and others. These structures are extraordinarily resistant to environmental stresses such as heat, ultraviolet radiation, gamma radiation, chemical disinfectants, and desiccation. In fact, some endospores have remained viable for around 100,000 years, and actinomycete spores (which are not true endospores) have been recovered alive after burial in the mud for 7,500 years. Because of their resistance and the fact that several species of endospore-forming bacteria are dangerous pathogens, endospores are of great practical importance in food, industrial, and medical microbiology. This is because it is essential to be able to sterilize solutions and solid objects. Endospores often survive boiling for an hour or more; therefore autoclaves (*see chapter 7*) must be used to sterilize many materials. Endospores are also of considerable theoretical interest. Because bacteria manufacture these intricate entities in a very organized fashion over a period of a few hours, spore formation is well suited for research on the construction of complex biological structures. In the environment, endospores aid in survival when moisture or nutrients are scarce. [Resistance of endospores to high temperature \(chapter 7\)](#)

Endospores can be examined with both light and electron microscopes. Because spores are impermeable to most stains, they often are seen as colorless areas in bacteria treated with methylene blue and other simple stains; special spore stains are used to make them clearly visible (*see chapter 2*). Spore position in the mother cell or **sporangium** frequently differs among species, making it of considerable value in identification. Spores

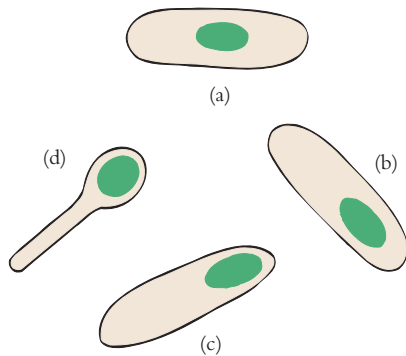


Figure 3.40 Examples of Endospore Location and Size. (a) Central spore. (b) Subterminal spore. (c) Terminal spore. (d) Terminal spore with swollen sporangium.

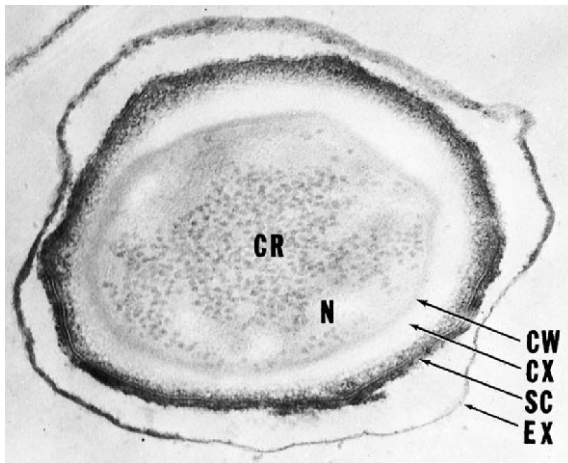


Figure 3.41 Endospore Structure. *Bacillus anthracis* endospore ($\times 151,000$). Note the following structures: exosporium, EX; spore coat, SC; cortex, CX; core wall, CW; and the protoplast or core with its nucleoid, N, and ribosomes, CR.

may be centrally located, close to one end (subterminal), or definitely terminal (**figure 3.40**). Sometimes a spore is so large that it swells the sporangium.

Electron micrographs show that endospore structure is complex (**figure 3.41**). The spore often is surrounded by a thin, delicate covering called the **exosporium**. A **spore coat** lies beneath the exosporium, is composed of several protein layers, and may be fairly thick. It is impermeable and responsible for the spore's resistance to chemicals. The **cortex**, which may occupy as much as half the spore volume, rests beneath the spore coat. It is made of a peptidoglycan that is less cross-linked than that in vegetative cells. The **spore cell wall** (or core wall) is inside the cortex and surrounds the protoplast or **core**. The core has the normal cell structures such as ribosomes and a nucleoid, but is metabolically inactive.

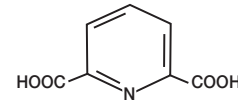


Figure 3.42 Dipicolinic Acid.

It is still not known precisely why the endospore is so resistant to heat and other lethal agents. As much as 15% of the spore's dry weight consists of **dipicolinic acid** complexed with calcium ions (**figure 3.42**), which is located in the core. It has long been thought that dipicolinic acid was directly involved in spore heat resistance, but heat-resistant mutants lacking dipicolinic acid now have been isolated. Calcium does aid in resistance to wet heat, oxidizing agents, and sometimes dry heat. It may be that calcium-dipicolinate often stabilizes spore nucleic acids. Recently specialized small, acid-soluble DNA-binding proteins have been discovered in the endospore. They saturate spore DNA and protect it from heat, radiation, desiccation, and chemicals. Dehydration of the protoplast appears to be very important in heat resistance. The cortex may osmotically remove water from the protoplast, thereby protecting it from both heat and radiation damage. The spore coat also seems to protect against enzymes and chemicals such as hydrogen peroxide. Finally, spores contain some DNA repair enzymes. DNA is repaired during germination and outgrowth after the core has become active once again. In summary, endospore heat resistance probably is due to several factors: calcium-dipicolinate and acid-soluble protein stabilization of DNA, protoplast dehydration, the spore coat, DNA repair, the greater stability of cell proteins in bacteria adapted to growth at high temperatures, and others.

Spore formation, **sporogenesis** or **sporulation**, normally commences when growth ceases due to lack of nutrients. It is a complex process and may be divided into seven stages (**figure 3.43**). An axial filament of nuclear material forms (stage I), followed by an inward folding of the cell membrane to enclose part of the DNA and produce the forespore septum (stage II). The membrane continues to grow and engulfs the immature spore in a second membrane (stage III). Next, cortex is laid down in the space between the two membranes, and both calcium and dipicolinic acid are accumulated (stage IV). Protein coats then are formed around the cortex (stage V), and maturation of the spore occurs (stage VI). Finally, lytic enzymes destroy the sporangium releasing the spore (stage VII). Sporulation requires only about 10 hours in *Bacillus megaterium*. [Regulation of sporulation in *Bacillus* \(pp. 282, 283–84\)](#)

The transformation of dormant spores into active vegetative cells seems almost as complex a process as sporogenesis. It occurs in three stages: (1) activation, (2) germination, and (3) outgrowth. Often an endospore will not germinate successfully, even in a nutrient-rich medium, unless it has been activated. Activation is a reversible process that prepares spores for

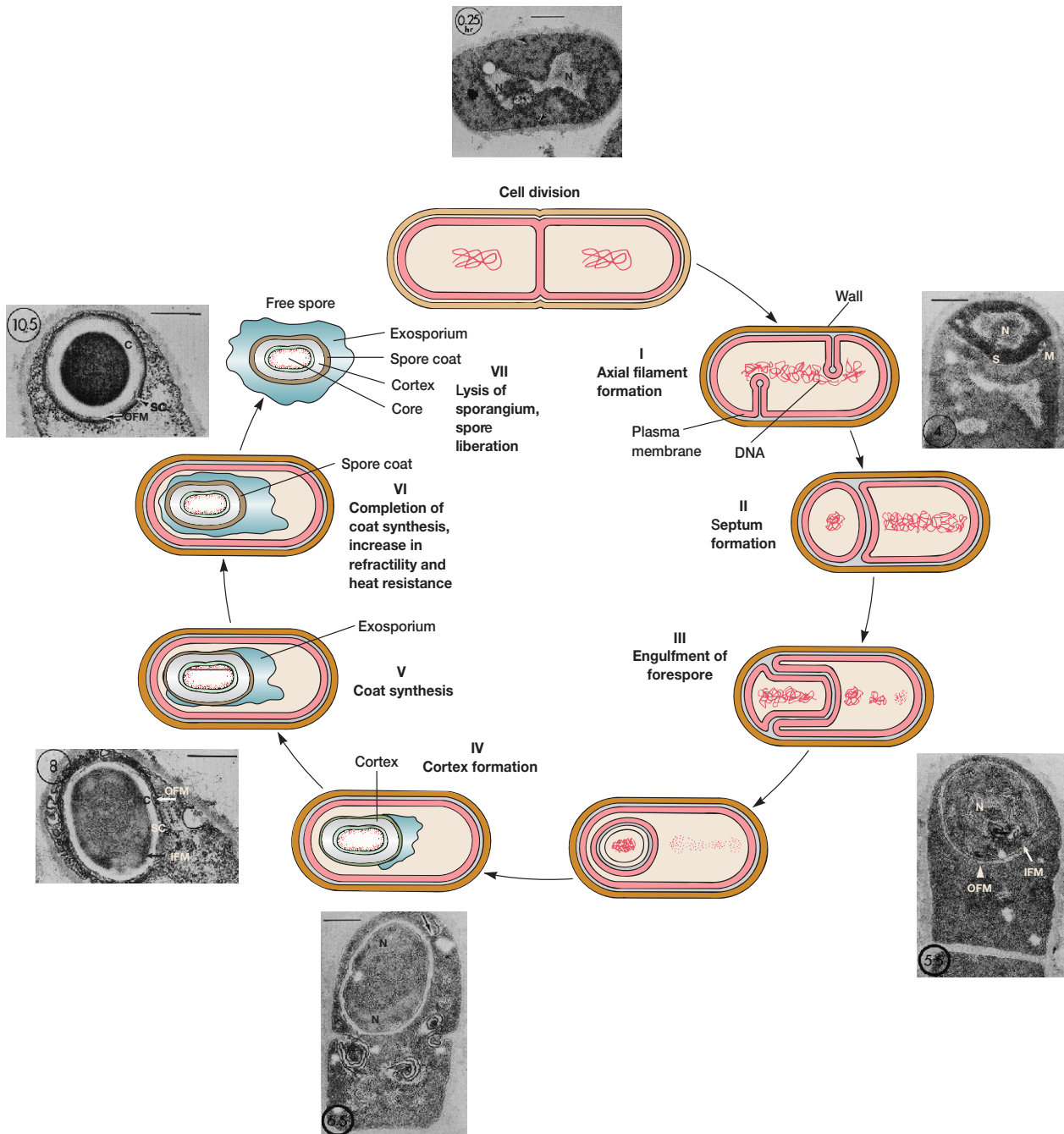


Figure 3.43 Endospore Formation: Life cycle of *Bacillus megaterium*. The stages are indicated by Roman numerals. The circled numbers in the photographs refer to the hours from the end of the logarithmic phase of growth: 0.25 h—a typical vegetative cell; 4 h—stage II cell, septation; 5.5 h—stage III cell, engulfment; 6.5 h—stage IV cell, cortex formation; 8 h—stage V cell, coat formation; 10.5 h—stage VI cell, mature spore in sporangium. Abbreviations used: C, cortex; IFM and OFM, inner and outer forespore membranes; M, mesosome; N, nucleoid; S, septum; SC, spore coats. Bars = 0.5 mm.

germination and usually results from treatments like heating. It is followed by **germination**, the breaking of the spore's dormant state. This process is characterized by spore swelling, rupture or absorption of the spore coat, loss of resistance to heat and other stresses, loss of refractility, release of spore components, and increase in metabolic activity. Many normal metabolites or nutrients (e.g., amino acids and sugars) can trigger germination after activation. Germination is followed by the third stage, outgrowth. The spore protoplast makes new components, emerges from the remains of the spore coat, and develops again into an active bacterium (**figure 3.44**).

1. Describe the structure of the bacterial endospore using a labeled diagram.
2. Briefly describe endospore formation and germination. What is the importance of the endospore? What might account for its heat resistance?

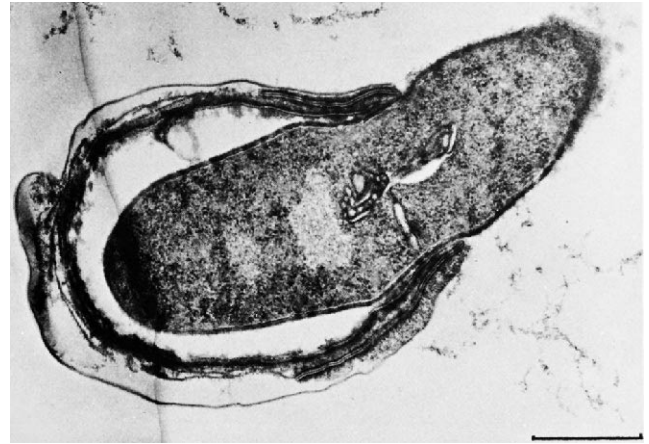


Figure 3.44 Endospore Germination. *Clostridium pectinovorum* emerging from the spore during germination. Bar = 0.5 μ m.

Summary

1. Bacteria may be spherical (cocci), rod-shaped (bacilli), spiral, or filamentous; form buds and stalks; or even have no characteristic shape at all (pleomorphic).
2. Bacterial cells can remain together after division to form pairs, chains, and clusters of various sizes and shapes.
3. All bacteria are prokaryotes and much simpler structurally than eukaryotes. **Table 3.1** summarizes the major functions of bacterial cell structures.
4. The plasma membrane and most other membranes are composed of a lipid bilayer in which integral proteins are buried (**figure 3.7**). Peripheral proteins are more loosely attached to membranes.
5. The plasma membrane may invaginate to form some simple structures such as membrane systems containing photosynthetic and respiratory assemblies, and possibly mesosomes.
6. The cytoplasmic matrix contains inclusion bodies and ribosomes.
7. Prokaryotic genetic material is located in an area called the nucleoid and is not enclosed by a membrane.
8. Most bacteria have a cell wall outside the plasma membrane to give them shape and protect them from osmotic lysis.
9. Bacterial walls are chemically complex and usually contain peptidoglycan or murein (**figures 3.16–3.19**).
10. Bacteria often are classified as either gram positive or gram negative based on differences in cell wall structure and their response to Gram staining.
11. Gram-positive walls have thick, homogeneous layers of peptidoglycan and teichoic acids (**figure 3.21**). Gram-negative bacteria have a thin peptidoglycan layer surrounded by a complex outer membrane containing lipopolysaccharides (LPSs) and other components (**figure 3.23**).
12. Bacteria such as mycoplasmas lack a cell wall.
13. Structures such as capsules, fimbriae, and sex pili are found outside the cell wall.
14. Many bacteria are motile, usually by means of threadlike locomotory organelles called flagella (**figure 3.32**).
15. Bacterial species differ in the number and distribution of their flagella.
16. The flagellar filament is a rigid helix and rotates like a propeller to push the bacterium through the water (**figures 3.35 and 3.36**).
17. Motile bacteria can respond to gradients of attractants and repellents, a phenomenon known as chemotaxis.
18. Some bacteria survive adverse environmental conditions by forming endospores, dormant structures resistant to heat, desiccation, and many chemicals (**figure 3.41**).

Key Terms

amphitrichous 63
axial filament 66
bacillus 43
basal body 64
capsule 61
carboxysomes 51
chemoreceptors 67
chemotaxis 67
coccus 42
core 69
core polysaccharide 58
cortex 69

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outer membrane 55

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peptide interbridge 56

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periplasm 55

periplasmic space 55

peritrichous 63

plasma membrane 46

plasmid 54

plasmolysis 61

pleomorphic 44

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poly- β -hydroxybutyrate (PHB) 49

polyphosphate granules 52

porin protein 60

protoplast 49

ribosome 52

rod 43

run 67

S-layer 62

self-assembly 65

sex pili 63

slime layer 61

spheroplast 61

spirilla 44

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teichoic acid 56

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Questions for Thought and Review

1. List all the major procaryotic structures discussed in this chapter and provide a brief description of the functions of each.
2. Some microbiologists believe that the plasma membrane is involved in DNA synthesis during bacterial reproduction. How might one prove that this is the case?
3. Discuss a possible mechanism of Gram staining in terms of differences in structure

- and chemistry between the walls of gram-positive and gram-negative bacteria.
4. What is self-assembly and why is it so important to cells?
5. How might one go about showing that a bacterium could form true endospores?

Critical Thinking Questions

1. Propose a model for the assembly of a flagellum in a gram-positive membrane. How would that model need to be modified for the assembly of a flagellum in a gram-negative membrane?
2. If you could not use a microscope, how would you determine whether a cell is procaryotic or eucaryotic? Assume the organism can be cultured easily in the laboratory.
3. The peptidoglycan of bacteria has been compared with the chain mail, worn beneath a medieval knight's suit of armor. It provides both protection and flexibility. Can you describe other structures in biology that have an analogous function? How are they replaced or modified to accommodate the growth of the inhabitant?

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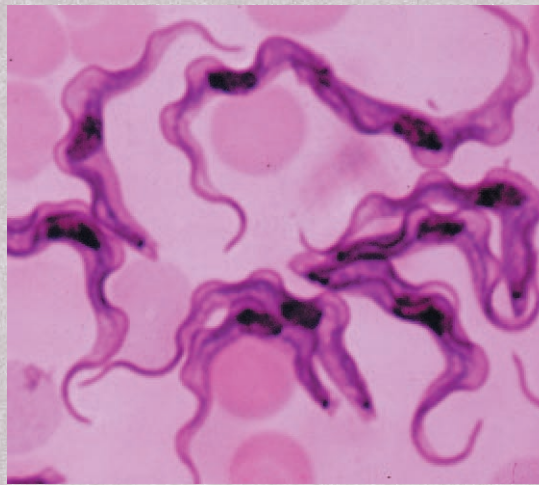
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CHAPTER 4

Eucaryotic Cell Structure and Function



Often we exclusively emphasize procaryotes and viruses, but eucaryotic microorganisms also have major impacts on human welfare. For example, the protozoan parasite *Trypanosoma brucei gambiense* is a cause of African sleeping sickness. The organism invades the nervous system and the victim frequently dies after suffering several years from symptoms such as weakness, headache, apathy, emaciation, sleepiness, and coma.

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Concepts

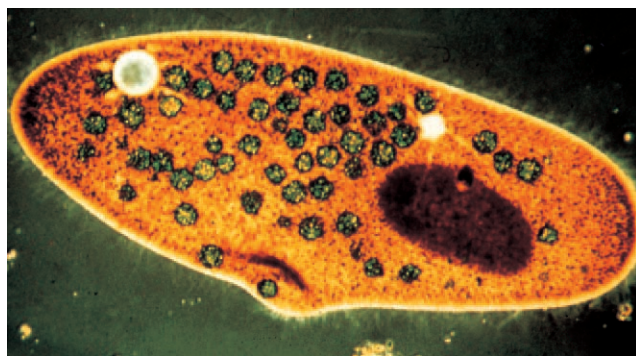
- 1. Eucaryotic cells differ most obviously from procaryotic cells in having a variety of complex membranous organelles in the cytoplasmic matrix and the majority of their genetic material within membrane-delimited nuclei. Each organelle has a distinctive structure directly related to specific functions.
- 2. A cytoskeleton composed of microtubules, microfilaments, and intermediate filaments helps give eucaryotic cells shape; microtubules and microfilaments are also involved in cell movements and intracellular transport.
- 3. In eucaryotes, genetic material is distributed between cells by the highly organized, complex processes called mitosis and meiosis.
- 4. Despite great differences between eucaryotes and procaryotes with respect to such things as morphology, they are similar on the biochemical level.

The key to every biological problem must finally be sought in the cell.
—E. B. Wilson

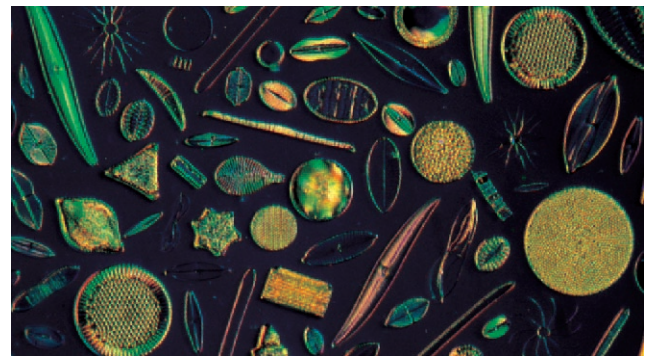
In chapter 3 considerable attention is devoted to procaryotic cell structure and function because bacteria are immensely important in microbiology and have occupied a large portion of microbiologists' attention in the past. Nevertheless, eucaryotic algae, fungi, and protozoa also are microorganisms and have been extensively studied. These organisms often are extraordinarily complex, interesting in their own right, and prominent members of the ecosystem (**figure 4.1**). In addition, fungi

(and to some extent, algae) are exceptionally useful in industrial microbiology. Many fungi and protozoa are also major human pathogens; one only need think of either malaria or African sleeping sickness (see chapter opener) to appreciate the significance of eucaryotes in pathogenic microbiology. So, although this text emphasizes bacteria, eucaryotic microorganisms are discussed at many points.

Chapter 4 focuses on eucaryotic cell structure and its relationship to cell function. Because many valuable studies on eucaryotic cell ultrastructure have used organisms other than microorganisms, some work on nonmicrobial cells is presented. At the end of the chapter, procaryotic and eucaryotic cells are compared in some depth.



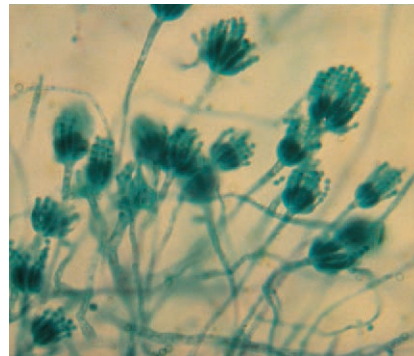
(a)



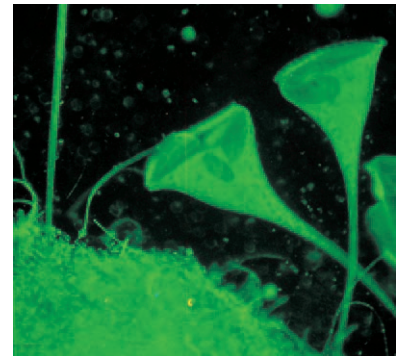
(b)



(c)



(d)



(e)



(f)

Figure 4.1 Representative Examples of Eucaryotic Microorganisms.

(a) *Paramecium* as seen with interference-contrast microscopy ($\times 115$). (b) Mixed diatom frustules ($\times 100$). (c) *Penicillium* colonies, and (d) a microscopic view of the mold's hyphae and conidia ($\times 220$). (e) *Stentor*. The ciliated protozoa are extended and actively feeding, dark-field microscopy ($\times 100$). (f) *Amanita muscaria*, a large poisonous mushroom ($\times 5$).

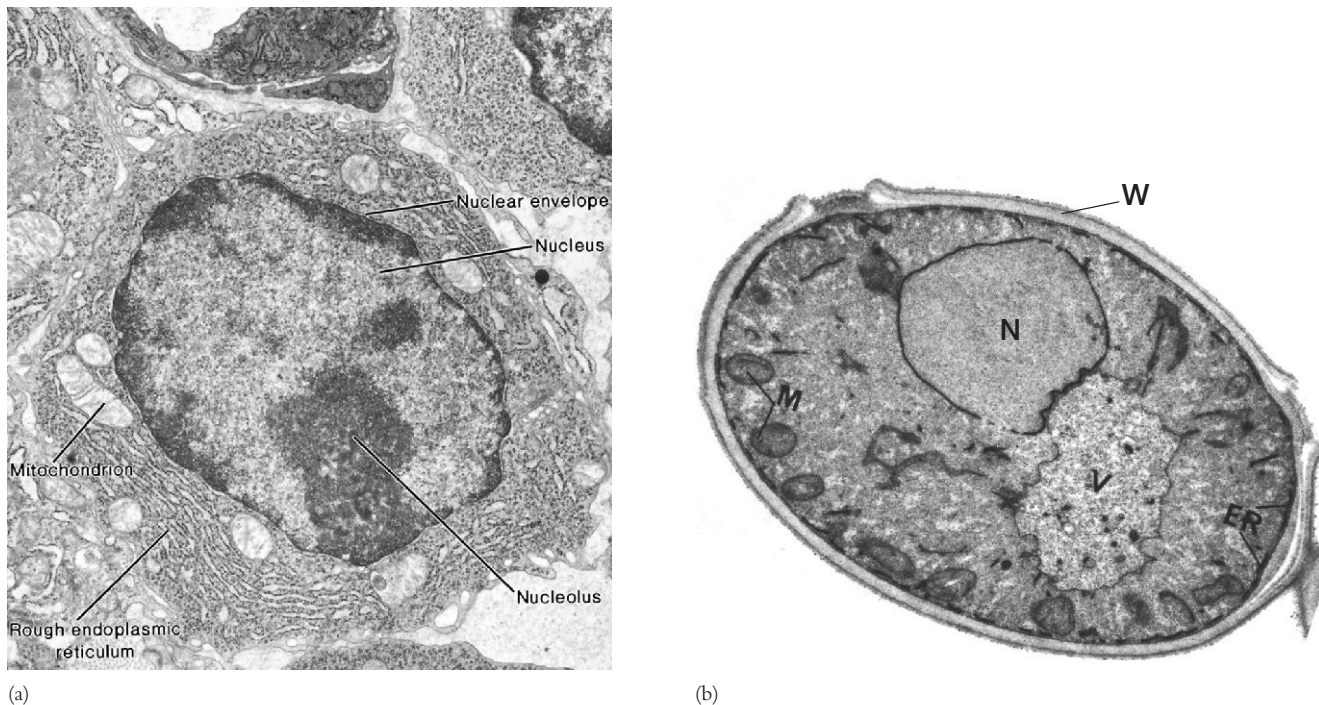


Figure 4.2 Eucaryotic Cell Ultrastructure. (a) A lymphoblast in the rat lymph node ($\times 17,500$). (b) The yeast *Saccharomyces* ($\times 7,200$). Note the nucleus (n), mitochondrion (m), vacuole (v), endoplasmic reticulum (er), and cell wall (w).

4.1 An Overview of Eucaryotic Cell Structure

The most obvious difference between eucaryotic and procaryotic cells is in their use of membranes. Eucaryotic cells have membrane-delimited nuclei, and membranes also play a prominent part in the structure of many other organelles (figures 4.2 and 4.3). **Organelles** are intracellular structures that perform specific functions in cells analogous to the functions of organs in the body. The name organelle (little organ) was coined because biologists saw a parallel between the relationship of organelles to a cell and that of organs to the whole body. It is not satisfactory to define organelles as membrane-bound structures because this would exclude such components as ribosomes and bacterial flagella. A comparison of figures 4.2 and 4.3 with figure 3.11 (p. 51) shows how much more structurally complex the eucaryotic cell is. This complexity is due chiefly to the use of internal membranes for several purposes. The partitioning of the eucaryotic cell interior by membranes makes possible the placement of different biochemical and physiological functions in separate compartments so that they can more easily take place simultaneously under independent control and proper coordination. Large membrane surfaces make possible greater respiratory and photosynthetic activity because these processes are located exclusively in membranes. The intracytoplasmic membrane complex also serves as a transport system to move materials be-

tween different cell locations. Thus abundant membrane systems probably are necessary in eucaryotic cells because of their large volume and the need for adequate regulation, metabolic activity, and transport.

Figures 4.2, 4.3, and 4.26b provide generalized views of eucaryotic cell structure and illustrate most of the organelles to be discussed. **Table 4.1** briefly summarizes the functions of the major eucaryotic organelles. Those organelles lying inside the plasma membrane are first described, and then components outside the membrane are discussed.

4.2 The Cytoplasmic Matrix, Microfilaments, Intermediate Filaments, and Microtubules

When a eucaryotic cell is examined at low power with the electron microscope, its larger organelles are seen to lie in an apparently featureless, homogeneous substance called the **cytoplasmic matrix**. The matrix, although superficially uninteresting, is actually one of the most important and complex parts of the cell. It is the “environment” of the organelles and the location of many important biochemical processes. Several physical changes seen in cells—viscosity changes, cytoplasmic streaming, and others—also are due to matrix activity.

Figure 4.3 Eucaryotic Cell Ultrastructure. This is a schematic, three-dimensional diagram of a cell with the most important organelles identified in the illustration. AV, autophagic vacuole; C, centriole; CH, chloroplast; CI, cilium; CR, chromatin; DV, digestion vacuole; F, microfilaments; G, glycogen; GA, Golgi apparatus; GE, GERL; LD, lipid droplet; M, mitochondrion; MT, microtubules; N, nucleus; NU, nucleolus; P, peroxisome; PL, primary lysosome; PM, plasma membrane; PV, pinocytotic vesicle; R, ribosomes and polysomes; RB, residual body; RER, rough endoplasmic reticulum; SER, smooth endoplasmic reticulum; SV, secretion vacuole.

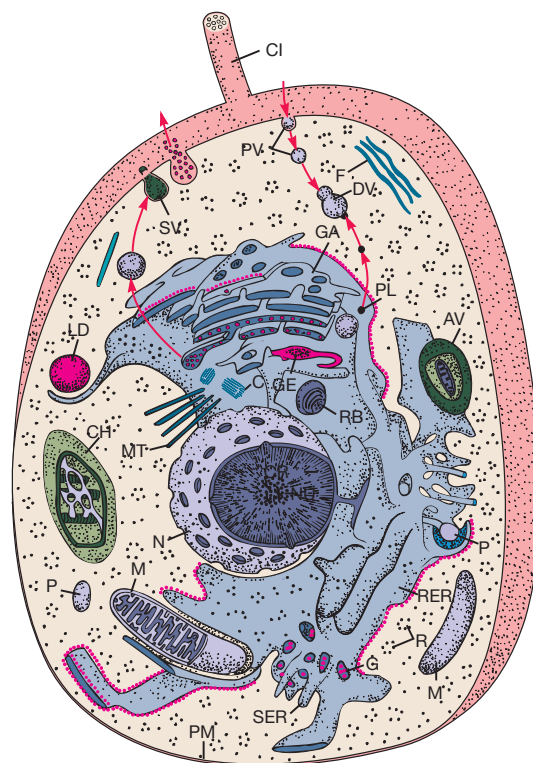


Table 4.1 Functions of Eucaryotic Organelles

Plasma membrane	Mechanical cell boundary, selectively permeable barrier with transport systems, mediates cell-cell interactions and adhesion to surfaces, secretion
Cytoplasmic matrix	Environment for other organelles, location of many metabolic processes
Microfilaments, intermediate filaments, and microtubules	Cell structure and movements, form the cytoskeleton
Endoplasmic reticulum	Transport of materials, protein and lipid synthesis
Ribosomes	Protein synthesis
Golgi apparatus	Packaging and secretion of materials for various purposes, lysosome formation
Lysosomes	Intracellular digestion
Mitochondria	Energy production through use of the tricarboxylic acid cycle, electron transport, oxidative phosphorylation, and other pathways
Chloroplasts	Photosynthesis—trapping light energy and formation of carbohydrate from CO ₂ and water
Nucleus	Repository for genetic information, control center for cell
Nucleolus	Ribosomal RNA synthesis, ribosome construction
Cell wall and pellicle	Strengthen and give shape to the cell
Cilia and flagella	Cell movement
Vacuole	Temporary storage and transport, digestion (food vacuoles), water balance (contractile vacuole)

Water constitutes about 70 to 85% by weight of a eucaryotic cell. Thus a large part of the cytoplasmic matrix is water. Cellular water can exist in two different forms. Some of it is bulk or free water; this is normal, osmotically active water. [Osmosis, water activity, and growth \(pp. 61, 121–23\)](#)

Water also can exist as bound water or water of hydration. This water is bound to the surface of proteins and other macromolecules and is osmotically inactive and more ordered than bulk water. There is some evidence that bound water is the site of many metabolic processes. The protein content of cells is so high that the cytoplasmic matrix often may be semicrystalline. Usually matrix pH is around neutrality, about pH 6.8 to 7.1, but can vary widely. For example, protozoan digestive vacuoles may reach pHs as low as 3 to 4.

Probably all eucaryotic cells have **microfilaments**, minute protein filaments, 4 to 7 nm in diameter, which may be either scattered within the cytoplasmic matrix or organized into networks and parallel arrays. Microfilaments are involved in cell motion and shape changes. Some examples of cellular movements associated with microfilament activity are the motion of pigment granules, amoeboid movement, and protoplasmic streaming in slime molds (*see chapter 25*).

The participation of microfilaments in cell movement is suggested by electron microscopic studies showing that they frequently are found at locations appropriate for such a role. For example, they are concentrated at the interface between stationary and flowing cytoplasm in plant cells and slime molds. Experiments using the drug cytochalasin B have provided additional